Paranasal sinuses

Maxillary, frontal, ethmoid and sphenoid sinuses: development, walls, drainage, the cells and variants that matter on CT, and the danger zones in FESS.

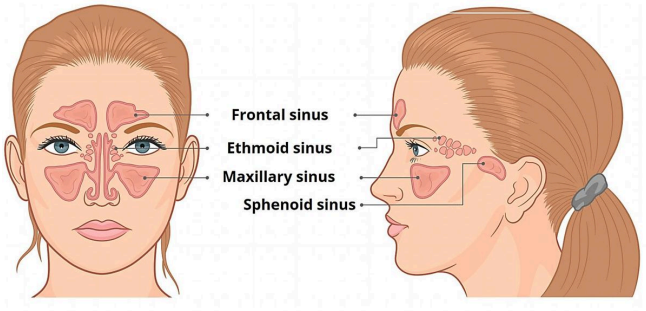
1. Overview and development
2. Maxillary sinus
3. Frontal sinus and frontal recess
4. Ethmoid bone and sinuses
5. Skull base dangers: Keros
6. Haller and Onodi cells
7. Sphenoid bone and sinus
8. Mucosa, drainage and supply
9. Pterygopalatine fossa
10. In theatre
11. On imaging

01 Overview and development

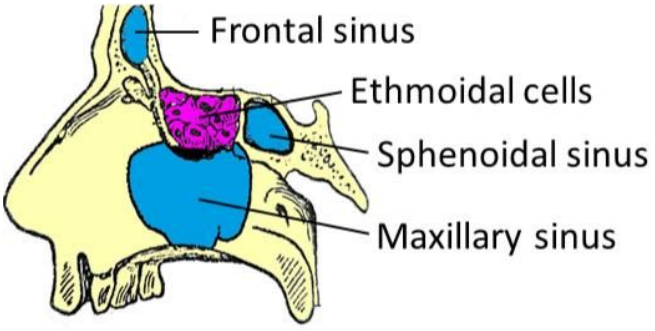
- Four paired sinuses named after their bones. **Anterior group** (maxillary, frontal, anterior ethmoid) drains in front of the basal lamella into the **middle meatus**. **Posterior group**: posterior ethmoid (superior meatus) and sphenoid (sphenoethmoidal recess).
- **Maxillary:** first to develop, present at birth; full size around 15 mL after the third molar erupts. Largest sinus.
- **Ethmoid:** present at birth.
- **Sphenoid:** absent at birth, recognisable around 3–6 years.
- **Frontal:** absent at birth, develops as an upward extension of the frontal recess; last to finish, often asymmetric or absent.

CLINICAL PEARL Frontal sinusitis is unusual before about 10 years; orbital complications in young children come from the ethmoids.

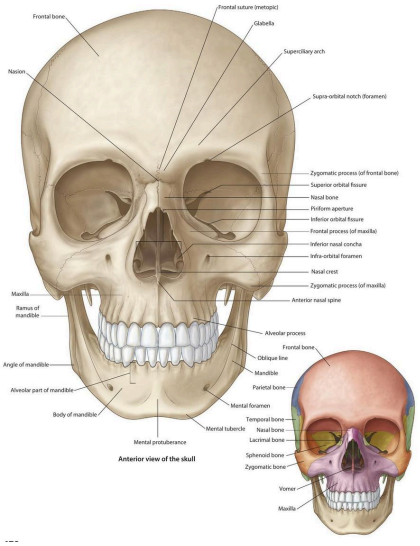
SOURCES DIFFER SPHENOID SINUS AT BIRTH: ENT Steps says absent, full size at 6 years; Dhingra and Bansal describe a tiny cavity at birth, visible on imaging at 3–4 years.



The four paired paranasal sinuses
ENT Steps Anatomy, p. 126

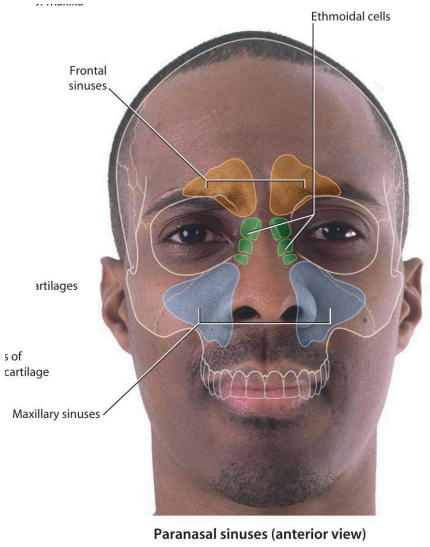


Paranasal sinuses in relation to the nasal cavity
Dr. Sameh Doss handout, p. 105



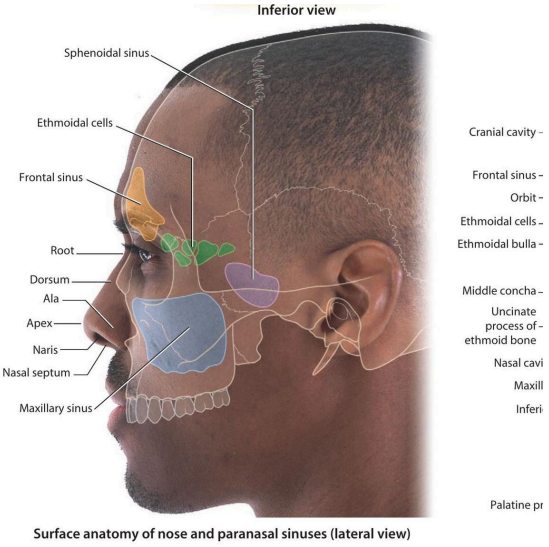
Skull, anterior view: orbits, piriform aperture and maxilla

Gray's Atlas of Anatomy, 2nd ed., p. 478



Paranasal sinuses (anterior view)

Gray's Atlas of Anatomy, 2nd ed., p. 566



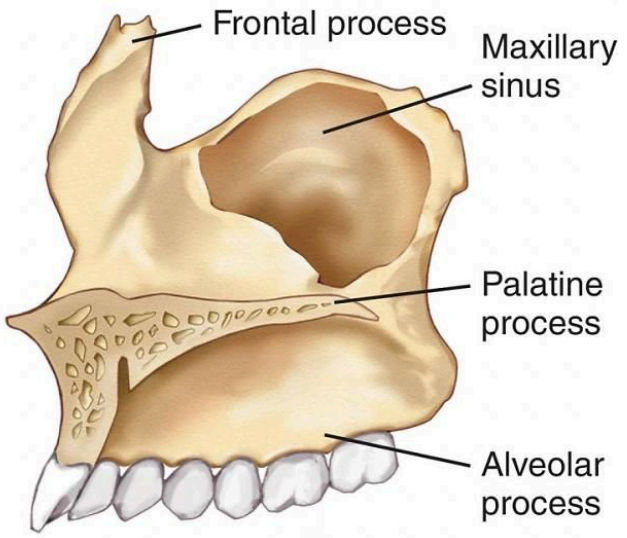
Surface anatomy of the nose and paranasal sinuses (lateral view)

Gray's Atlas of Anatomy, 2nd ed., p. 566

02 Maxillary sinus

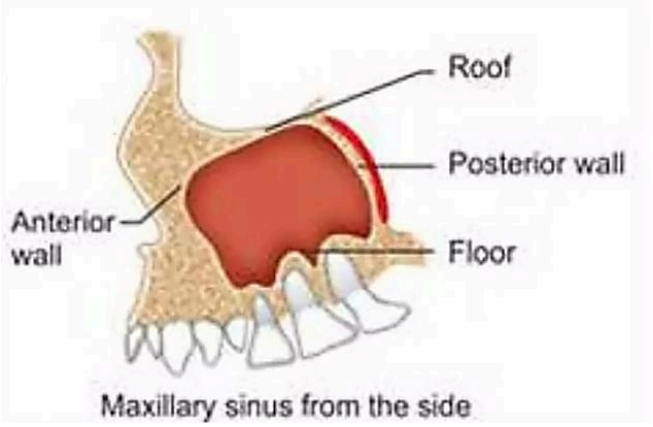
- Pyramid: **base** = lateral nasal wall; **apex** into the zygomatic process.
- **Anterior wall:** canine fossa (Caldwell–Luc approach). **Posterior wall:** infratemporal and pterygopalatine fossae. **Roof:** orbital floor with the **infraorbital nerve** in its canal.
- **Floor:** alveolar process, about **1 cm below the nasal floor** in adults. Roots of the **molars and second premolar** lie close to or in it.
- **Ostium:** high on the medial wall, just below the roof, opening into the lowest part of the ethmoidal infundibulum, so drainage relies on cilia, not gravity.
- Accessory ostium in about 30%, in the posterior fontanelle.

> **CLINICAL PEARL** Dental extraction of an upper molar can leave an oroantral fistula; odontogenic infection causes unilateral maxillary sinusitis.



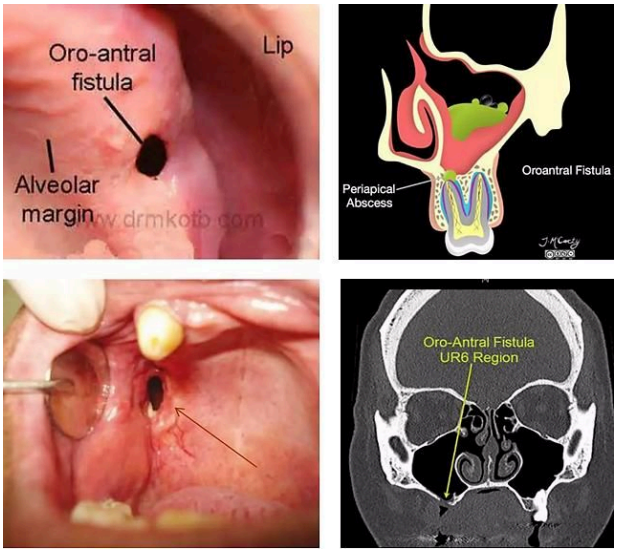
Maxilla and maxillary sinus, lateral view

ENT Steps Anatomy, p. 129



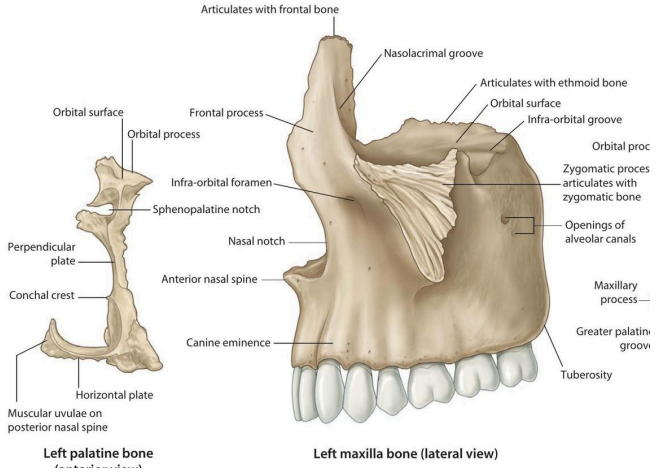
Walls of the maxillary sinus

ENT Steps Anatomy, p. 131



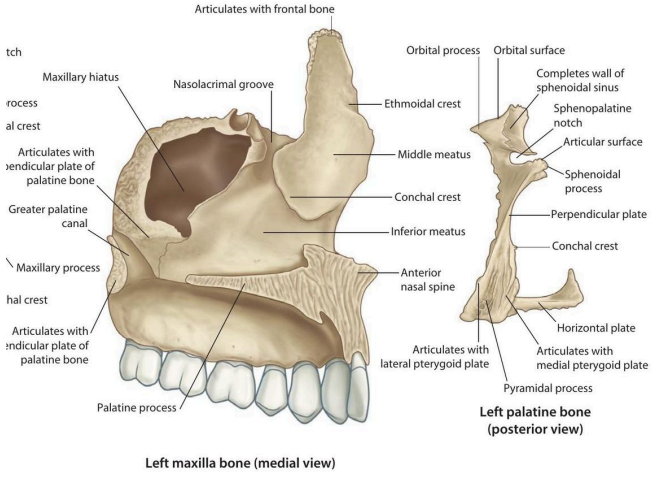
Oroantral fistula: relation of the maxillary sinus floor to the teeth

ENT Steps Anatomy, p. 132



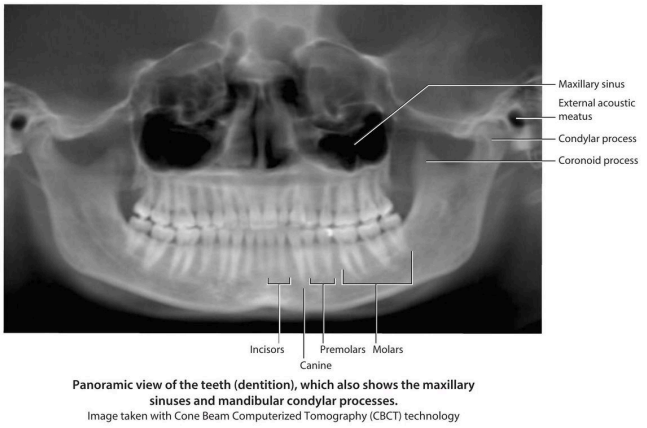
Left maxilla (lateral view)

Gray's Atlas of Anatomy, 2nd ed., p. 487



Left maxilla (medial view): maxillary hiatus

Gray's Atlas of Anatomy, 2nd ed., p. 487



Panoramic view of the teeth (dentition), which also shows the maxillary sinuses and mandibular condylar processes.
Image taken with Cone Beam Computerized Tomography (CBCT) technology

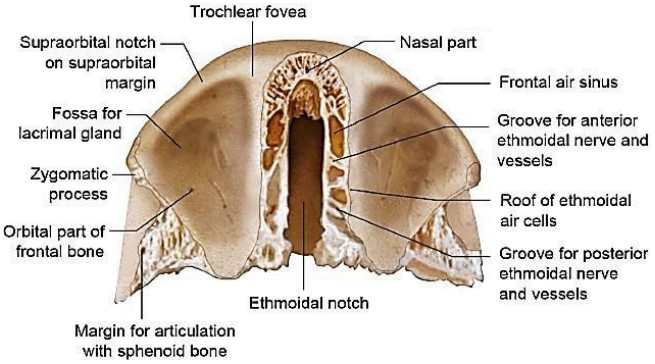
Panoramic radiograph: teeth and maxillary sinuses

Gray's Atlas of Anatomy, 2nd ed., p. 574

03 Frontal sinus and frontal recess

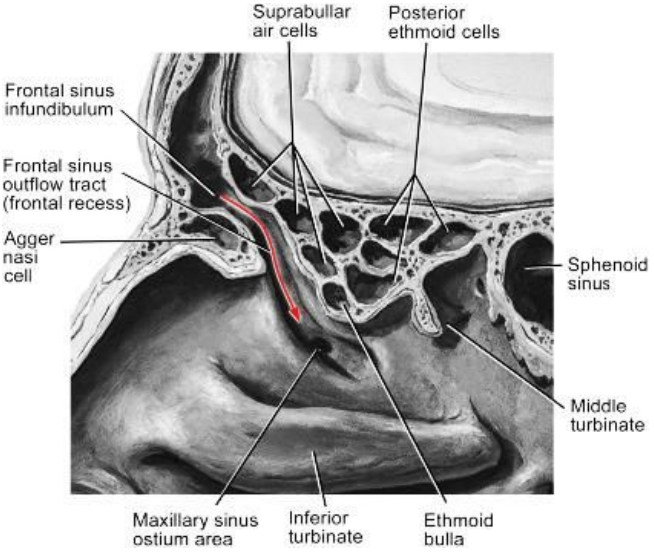
- Between the inner and outer tables above the orbit. **Anterior wall** thick (forehead); **floor** thinnest (orbital roof); **posterior wall** against the dura and frontal lobe.
- Drains through its floor into the **frontal recess**, then into the infundibulum or medial to the uncinate into the middle meatus.
- **Agger nasi cell:** most anterior ethmoid cell, in front of the middle turbinate attachment; it forms the anterior boundary of the frontal recess.
- **Frontal cells (Bent–Kuhn):** type I single cell above the agger nasi; type II tier of cells; type III single cell extending into the frontal sinus; type IV isolated cell within the sinus.

> **CLINICAL PEARL** Pott's puffy tumour: frontal osteomyelitis with subperiosteal abscess; infection through the thin posterior wall gives epidural or brain abscess.



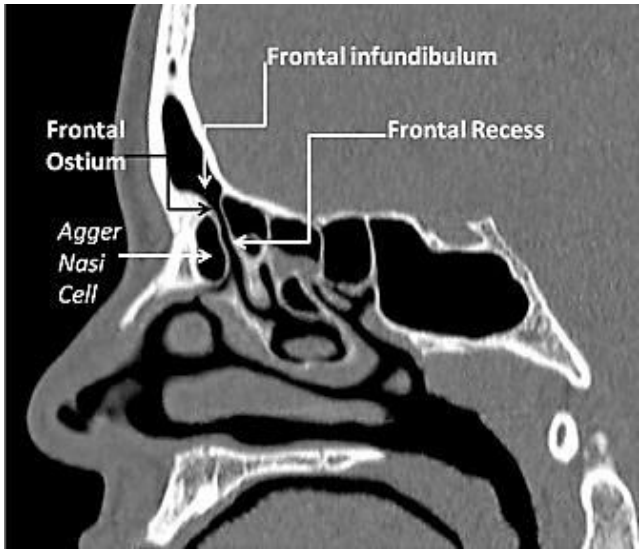
Frontal bone with the frontal sinus

ENT Steps Anatomy, p. 136



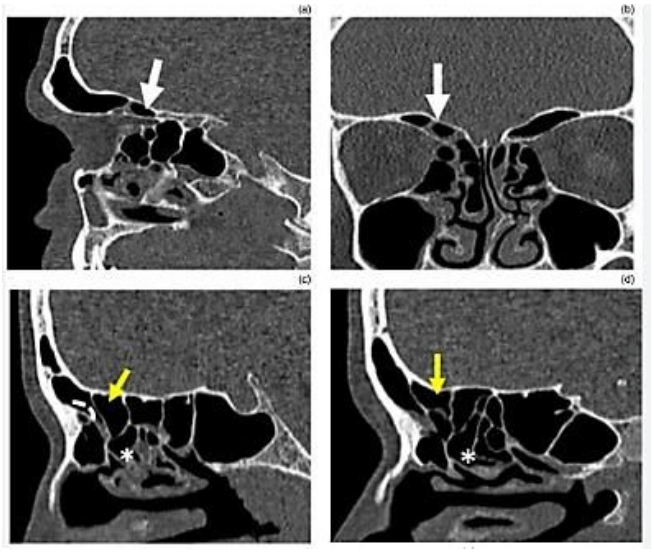
Frontal sinus drainage pathway and agger nasi cell

ENT Steps Anatomy, p. 138



Coronal CT: frontal recess and agger nasi cell

ENT Steps Anatomy, p. 138



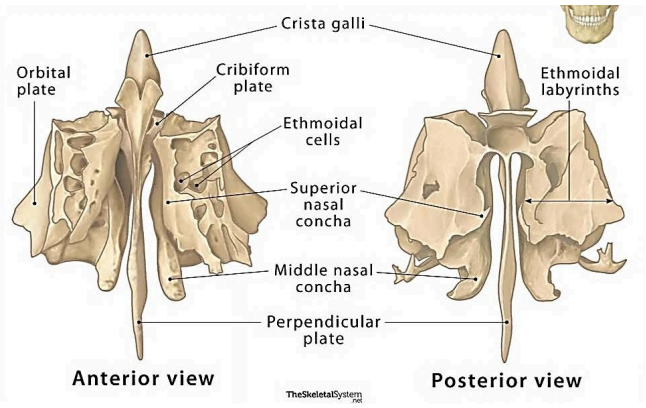
Frontal cells (Bent and Kuhn)

ENT Steps Anatomy, p. 144

04 Ethmoid bone and sinuses

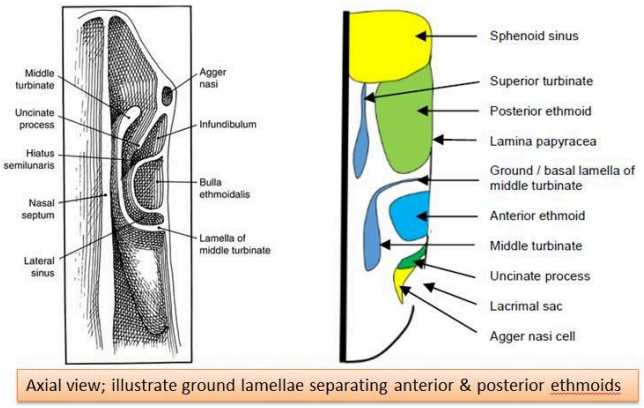
- 3–18 thin-walled cells between the upper lateral nasal wall and the medial orbital wall.
- **Lateral wall: lamina papyracea** (paper-thin) with the orbit beyond. **Roof: fovea ethmoidalis** of the frontal bone.
- **Basal lamella of the middle turbinate** divides **anterior cells** (open into the middle meatus) from **posterior cells** (open into the superior meatus).
- **Anterior ethmoidal artery** crosses the roof just behind the frontal recess, often in a bony canal or a mesentery below the skull base. It passes through orbit, ethmoid and anterior cranial fossa.
- **Supraorbital ethmoid cells** pneumatise over the orbit and can be mistaken for the frontal sinus.

CLINICAL PEARL Cutting the anterior ethmoidal artery lets it retract into the orbit and causes a rapidly expanding orbital haematoma. Lateral canthotomy is the emergency treatment.



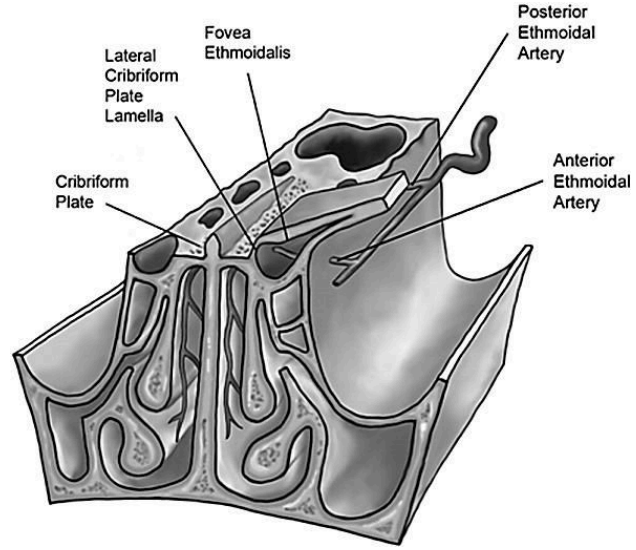
Ethmoid bone: anterior and posterior views

ENT Steps Anatomy, p. 140



Ground lamellae of the ethmoid separating anterior and posterior cells

ENT Steps Anatomy, p. 142



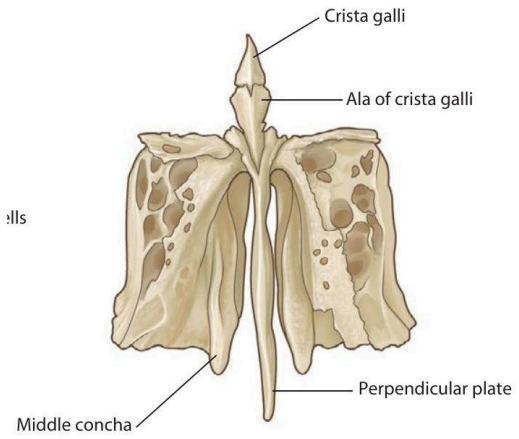
Ethmoid roof and anterior ethmoidal artery

ENT Steps Anatomy, p. 139



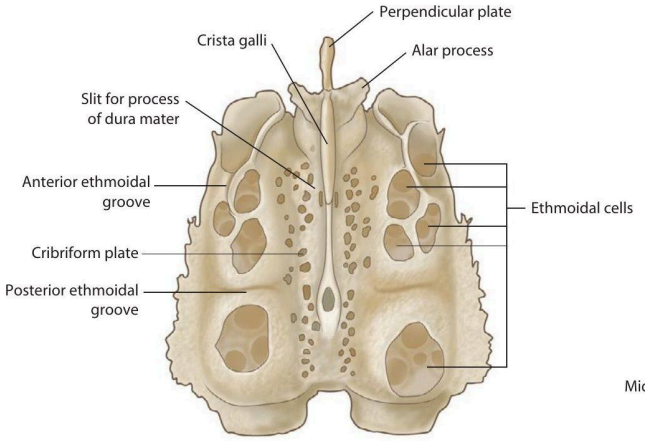
Supraorbital ethmoid cells

ENT Steps Anatomy, p. 145



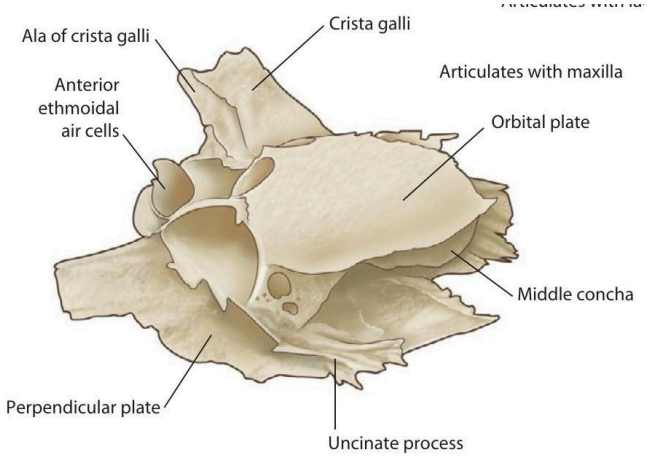
Ethmoid bone (posterior view)

Ethmoid bone (posterior view)
Gray's Atlas of Anatomy, 2nd ed., p. 486



Ethmoid bone (superior view)

Ethmoid bone (superior view): crista galli and cribriform plate
Gray's Atlas of Anatomy, 2nd ed., p. 486



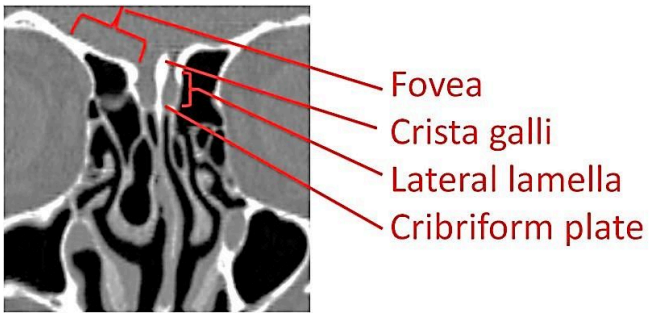
Ethmoid bone (lateral view)

Ethmoid bone (lateral view): uncinate process, middle concha, orbital plate
Gray's Atlas of Anatomy, 2nd ed., p. 486

05 Skull base dangers: Keros

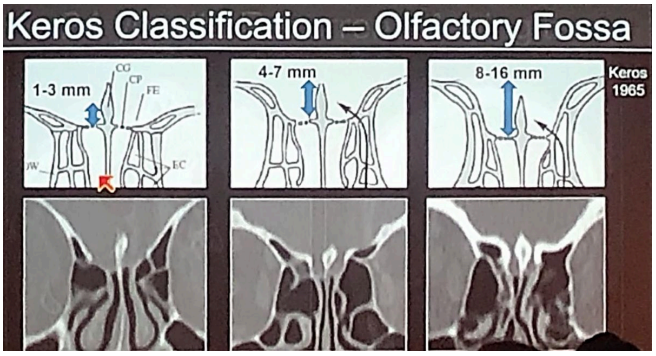
- The **lateral lamella of the cribriform plate** is the thinnest bone of the skull base and the commonest site of CSF leak in FESS.
- **Keros classification** (depth of the olfactory fossa below the fovea): **type I 1–3 mm, type II 4–7 mm, type III 8–16 mm**.
- Type III has the tallest, thinnest lateral lamella and the highest risk; asymmetry between sides adds risk.

> **CLINICAL PEARL** Always read the coronal CT for Keros type and asymmetry before ethmoidectomy.



Coronal CT: fovea ethmoidalis, lateral lamella and cribriform plate

ENT Steps Anatomy, p. 139



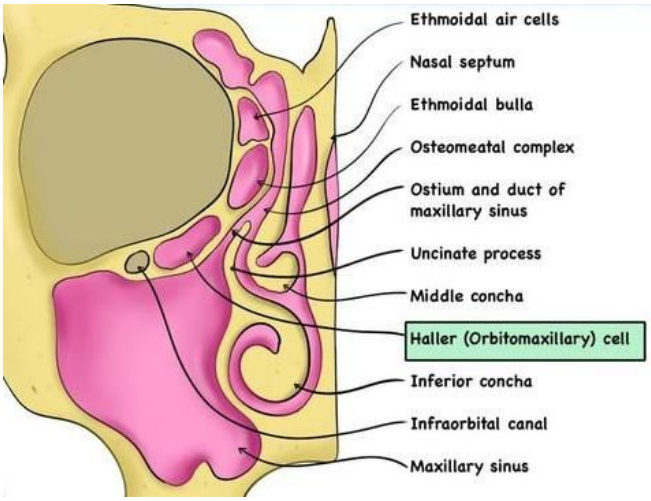
Keros classification of the olfactory fossa

ENT Steps Anatomy, p. 143

06 Haller and Onodi cells

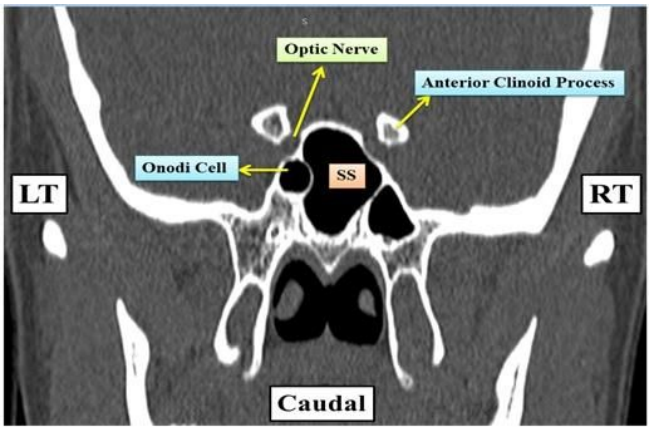
- **Haller (infraorbital ethmoid) cell:** extends along the orbital floor above the maxillary ostium; reported in roughly 10–45% depending on the series. It can narrow the infundibulum and cause recurrent maxillary sinusitis.
- **Onodi (sphenoethmoidal) cell:** posterior ethmoid cell that pneumatises lateral or above the sphenoid sinus; the **optic nerve** (and sometimes the ICA) can lie in its wall.

CLINICAL PEARL If you think you are in the sphenoid but see the optic nerve bulge in the lateral wall, you may be in an Onodi cell.



Haller (infraorbital ethmoid) cell

ENT Steps Anatomy, p. 146



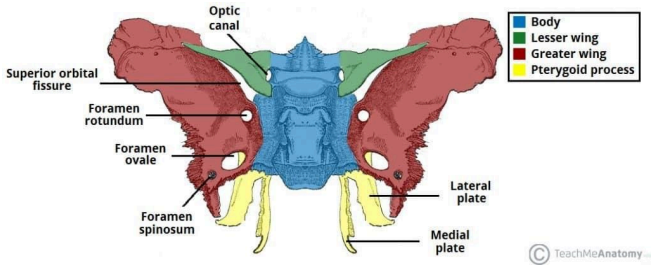
Coronal CT: Onodi cell around the optic nerve

ENT Steps Anatomy, p. 146

07 Sphenoid bone and sinus

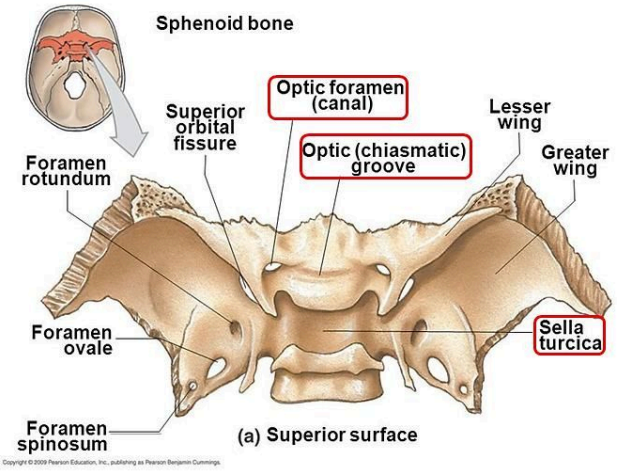
- Sinuses in the body of the sphenoid, separated by an often **oblique, off-midline septum**.
- **Pneumatisation: sellar** (most common, about 80%), **presellar** (about 17%), **conchal** (about 3%; makes transsphenoidal surgery hard).
- **Lateral wall: optic nerve** above, **ICA** below it (opticocarotid recess between them), **cavernous sinus** with III, IV, V1, V2, VI; maxillary nerve lower down. These may be **dehiscent**.
- **Roof:** sella turcica and pituitary; optic chiasm and olfactory tract in front.
- **Floor: vidian nerve** in the pterygoid canal.
- **Ostium:** upper anterior wall into the **sphenoethmoidal recess**, about 1–1.5 cm above the choana, medial to the superior turbinate.

CLINICAL PEARL The intersinus septum often inserts onto the ICA canal; twisting it off can tear the carotid.



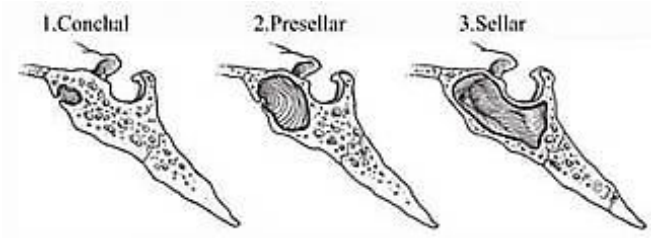
Sphenoid bone: body, wings and pterygoid processes

ENT Steps Anatomy, p. 147



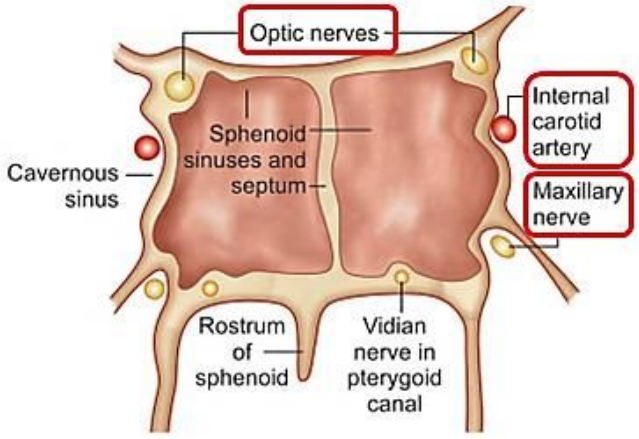
Sphenoid bone, superior surface

ENT Steps Anatomy, p. 148



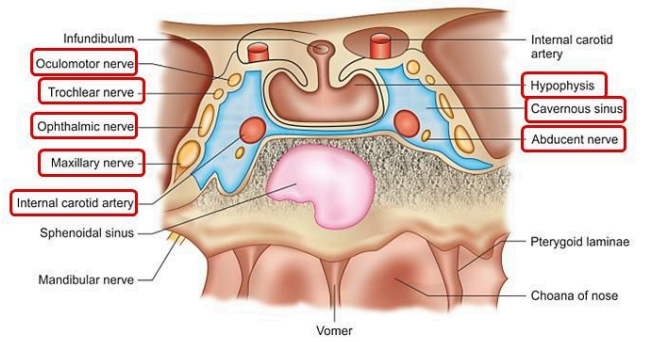
Sphenoid pneumatization: conchal, presellar, sellar

ENT Steps Anatomy, p. 153



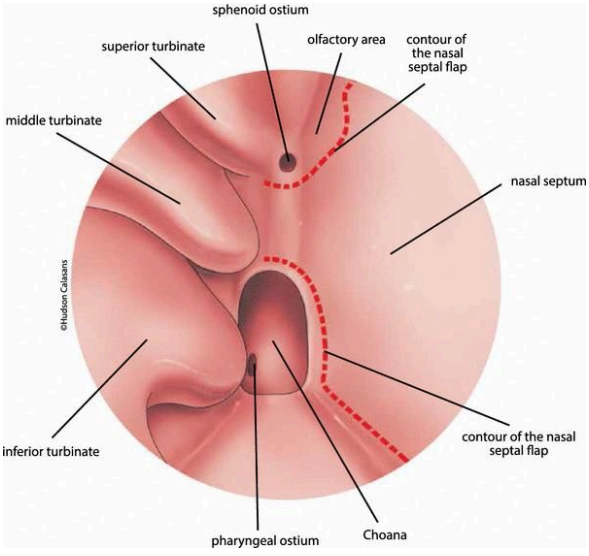
Relations of the sphenoid sinus: optic nerve, ICA, vidian nerve

ENT Steps Anatomy, p. 154



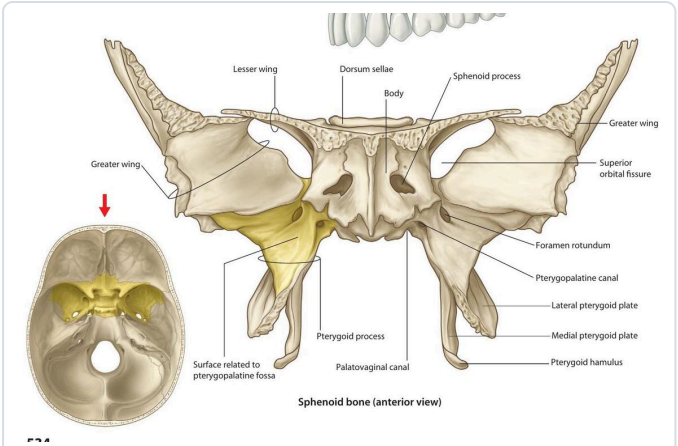
Coronal section: cavernous sinus beside the sphenoid

ENT Steps Anatomy, p. 155



Endoscopic view of the sphenoethmoidal recess

ENT Steps Anatomy, p. 156



Sphenoid bone (anterior view)

Gray's Atlas of Anatomy, 2nd ed., p. 534

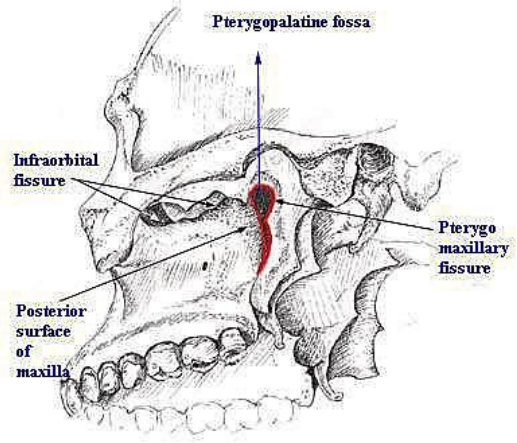
08 Mucosa, drainage and supply

- Ciliated pseudostratified columnar epithelium with goblet cells; thinner and less vascular than nasal mucosa. Mucus has a serous sol layer and a viscous gel layer.
- Cilia move mucus toward the **natural ostium** in set patterns: in the maxillary sinus a stellate pattern from the floor up to the ostium; in the frontal sinus up the medial side and out along the floor.
- **Arteries:** ECA (sphenopalatine, facial) and ICA (anterior and posterior ethmoidal).
- **Nerves:** V1 (supraorbital, supratrochlear, ethmoidal) and V2 (infraorbital, greater palatine, posterolateral nasal). Lymph to the retropharyngeal and upper deep cervical nodes.

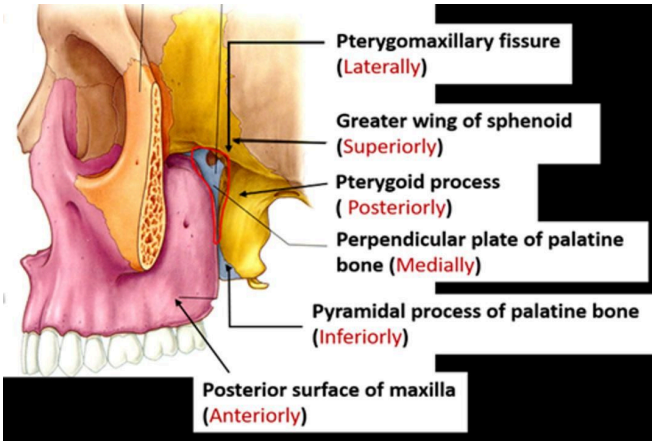
09 Pterygopalatine fossa

- Behind the maxillary sinus. Roof: sphenoid body; front: maxilla; back: pterygoid plates; medial: perpendicular plate of palatine.
- Contains the **maxillary artery (3rd part)**, **V2** and the **pterygopalatine ganglion**.
- Opens to the orbit, middle cranial fossa, foramen lacerum, nasopharynx, nasal cavity (**sphenopalatine foramen**), infratemporal fossa (**pterygomaxillary fissure**) and the mouth (palatine canals).

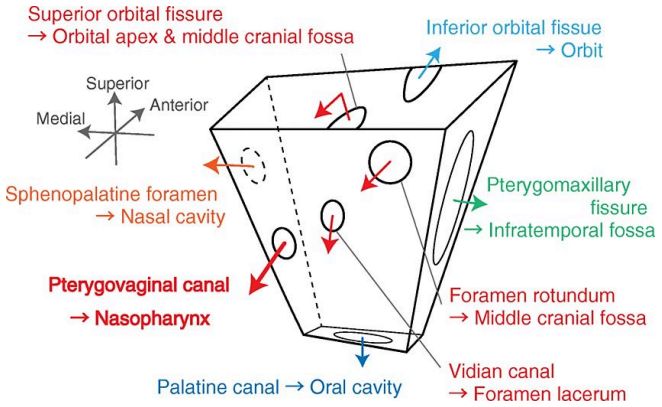
> **CLINICAL PEARL** Endoscopic access to the fossa is through the posterior wall of the maxillary sinus after a wide antrostomy.



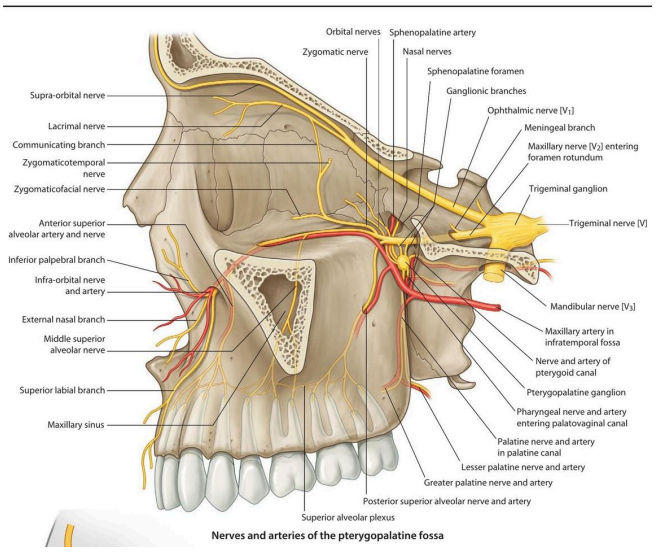
Pterygopalatine fossa, lateral view
ENT Steps Anatomy, p. 160



Boundaries of the pterygopalatine fossa
ENT Steps Anatomy, p. 160



Communications of the pterygopalatine fossa
ENT Steps Anatomy, p. 161

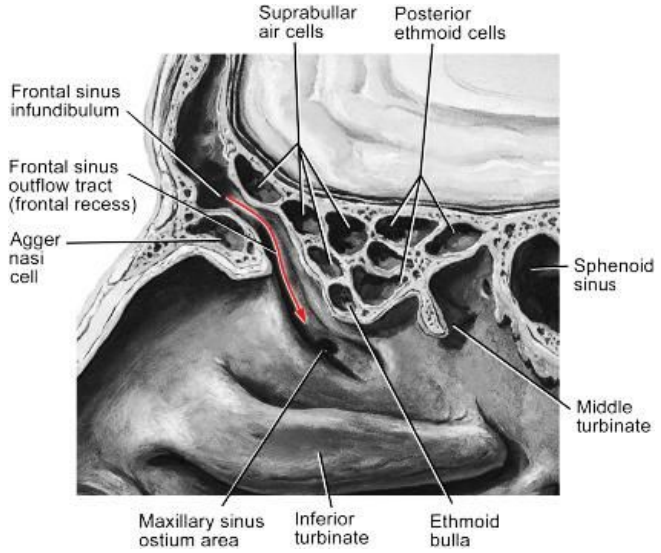


Nerves and arteries of the pterygopalatine fossa
Gray's Atlas of Anatomy, 2nd ed., p. 535

10 Surgical anatomy: In theatre

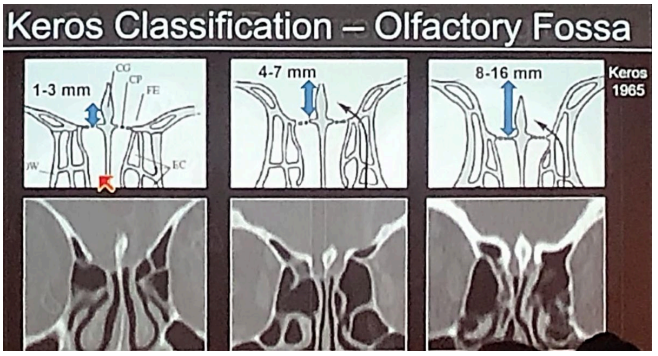
- **FESS steps:** uncinectomy → maxillary antrostomy (enlarge the natural ostium posteriorly and inferiorly) → bulla ethmoidalis → basal lamella → posterior ethmoids → sphenoidotomy → frontal recess.
- **Find the skull base** in the posterior ethmoid or sphenoid where it is highest and work from posterior to anterior; at the roof stay lateral (fovea ethmoidalis) and avoid instrumenting medially toward the lateral lamella of the cribriform plate.
- **Sphenoid ostium:** medial to the superior turbinate, about 1–1.5 cm above the choana, at about 30° from the nasal floor; enter low and medial.
- **Frontal recess:** remove the agger nasi and frontal cells; the anterior ethmoidal artery sits behind the frontal recess at the skull base.
- **Dangers:** orbit (lamina papyracea), anterior ethmoidal artery, skull base (lateral lamella, Keros III), optic nerve and ICA (sphenoid lateral wall, Onodi cell).

> **CLINICAL PEARL** Use the CLOSE checklist on the CT before FESS: Cribriform plate (Keros), Lamina papyracea, Onodi cell, Sphenoid (pneumatisation, septum on ICA), Ethmoidal artery.



Frontal sinus drainage pathway and agger nasi cell

ENT Steps Anatomy, p. 138

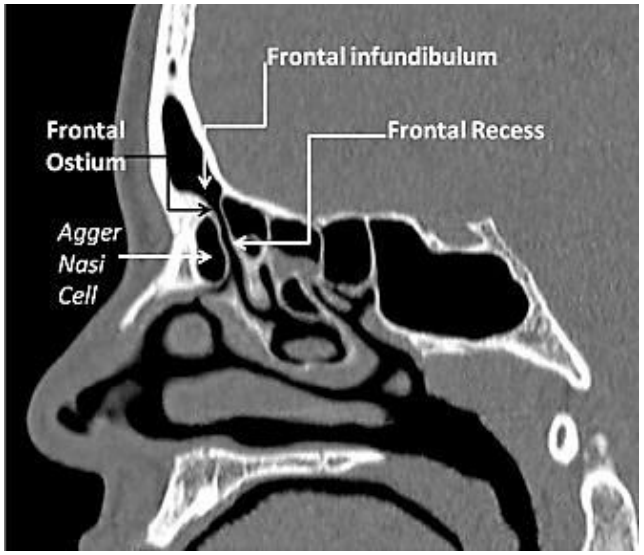


Keros classification of the olfactory fossa

ENT Steps Anatomy, p. 143

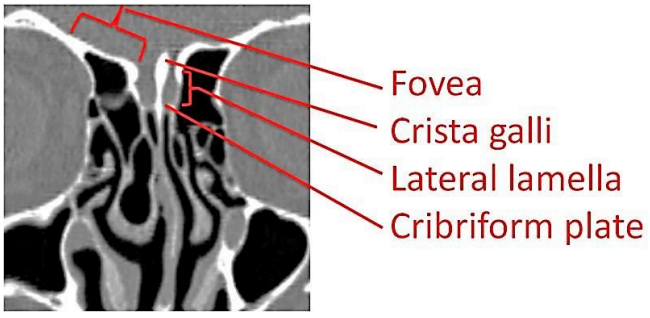
11 Radiological anatomy: On imaging

- **Non-contrast CT sinus**, coronal, axial and sagittal, bone window, is the roadmap for surgery; scan after medical treatment.
- **Lund–Mackay score:** each sinus 0 (clear), 1 (partial), 2 (complete); OMC 0 or 2; total out of 24.
- **MRI** for suspected tumour, fungal disease (signal void on T2), and intracranial or orbital spread.
- Fungal ball: dense calcified focus in a single sinus, usually the maxillary. Allergic fungal sinusitis: expanded sinuses with hyperdense, heterogeneous contents.



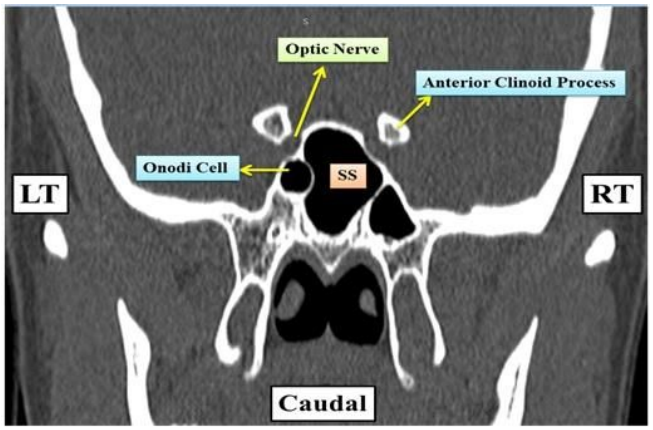
Coronal CT: frontal recess and agger nasi cell

ENT Steps Anatomy, p. 138



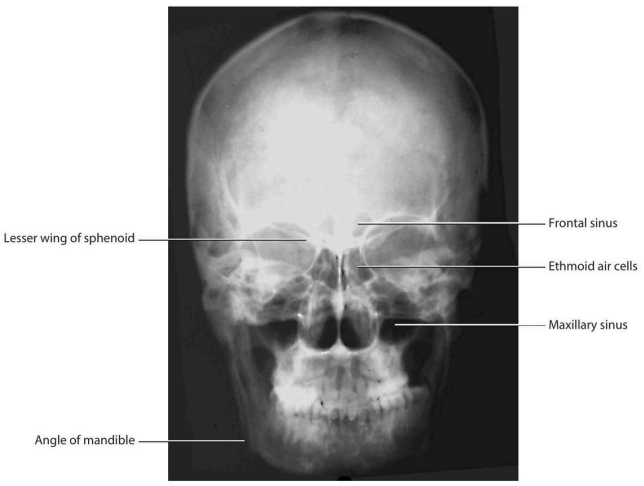
Coronal CT: fovea ethmoidalis, lateral lamella and cribriform plate

ENT Steps Anatomy, p. 139



Coronal CT: Onodi cell around the optic nerve

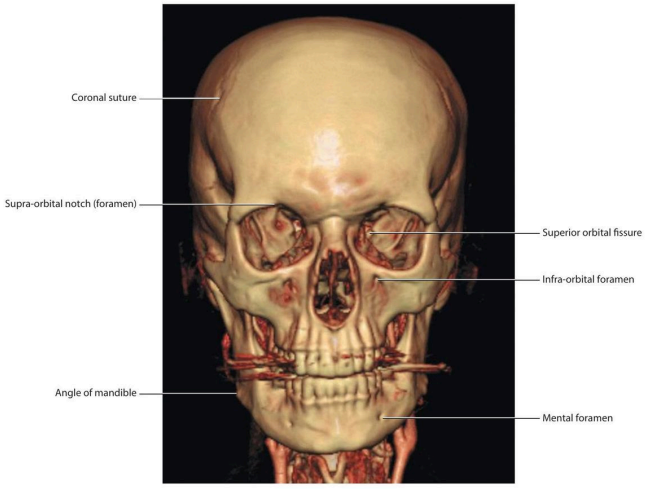
ENT Steps Anatomy, p. 146



Anterior view of the skull.
Radiograph, AP view

Skull radiograph, AP view: frontal, ethmoid and maxillary sinuses

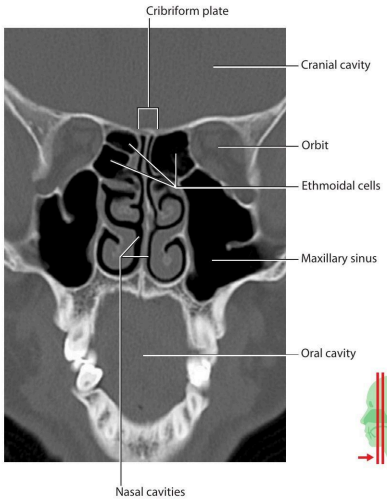
Gray's Atlas of Anatomy, 2nd ed., p. 479



Anterior view of the skull.
Volume-rendered anterior view using multidetector computed tomography

Skull, anterior view: volume-rendered CT

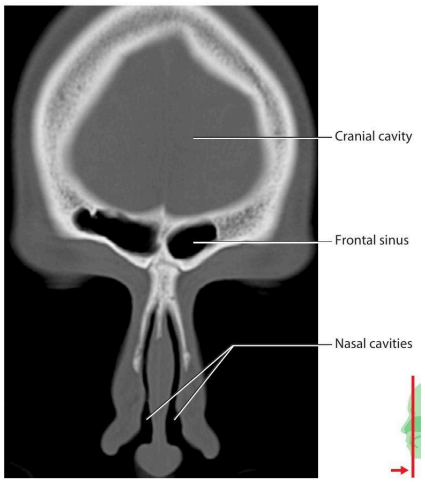
Gray's Atlas of Anatomy, 2nd ed., p. 479



Anterior view of the maxillary sinus and ethmoidal cells.
CT image in coronal plane

Coronal CT: maxillary sinus and ethmoidal cells

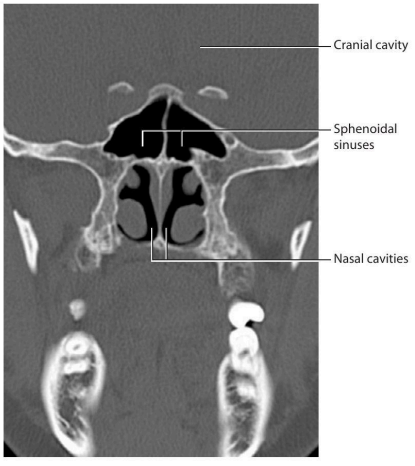
Gray's Atlas of Anatomy, 2nd ed., p. 570



Anterior view of the frontal sinuses.
CT image in coronal plane

Coronal CT: frontal sinuses

Gray's Atlas of Anatomy, 2nd ed., p. 570



Anterior view of the sphenoidal sinuses showing their relationship to the nasal cavity.
CT image in coronal plane

Coronal CT: sphenoidal sinuses

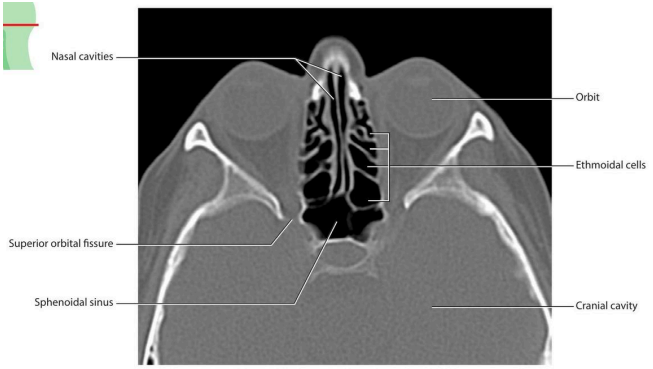
Gray's Atlas of Anatomy, 2nd ed., p. 570



Axial section showing the relationship among the nasal cavity, nasopharynx, and maxillary sinuses.
CT image in axial plane

Axial CT: nasal cavity, nasopharynx and maxillary sinuses

Gray's Atlas of Anatomy, 2nd ed., p. 571



Axial section showing the ethmoidal cells and the sphenoidal sinuses and the relationship of these structures to the orbit.
CT image in axial plane

Axial CT: ethmoidal cells, sphenoidal sinuses and orbit

Gray's Atlas of Anatomy, 2nd ed., p. 571

Tables

Skull base foramina and their contents

Foramen	Bone / fossa	Contents
Cribriform plate	Ethmoid · anterior fossa	Olfactory nerves (I), anterior ethmoidal nerve and vessels
Optic canal	Lesser wing of sphenoid	Optic nerve (II), ophthalmic artery
Superior orbital fissure	Between greater and lesser wings	III, IV, VI, V1 (lacrimal, frontal, nasociliary), superior ophthalmic vein
Foramen rotundum	Greater wing · middle fossa	Maxillary nerve (V2)
Foramen ovale	Greater wing · middle fossa	V3, accessory meningeal artery, lesser petrosal nerve, emissary vein (MALE)
Foramen spinosum	Greater wing · middle fossa	Middle meningeal artery and vein, meningeal branch of V3
Foramen lacerum	Between sphenoid, petrous and occipital	Cartilage; ICA crosses its upper part; greater + deep petrosal form the vidian nerve
Internal acoustic meatus	Petrous · posterior fossa	VII, nervus intermedius, VIII, labyrinthine artery and vein
Jugular foramen	Petrous + occipital · posterior fossa	IX, X, XI, IJV (sigmoid sinus), inferior petrosal sinus
Hypoglossal canal	Occipital	XII, meningeal branch of ascending pharyngeal artery
Stylomastoid foramen	Temporal	Facial nerve (VII), stylomastoid artery
Petrotympanic fissure	Temporal	Chorda tympani, anterior tympanic artery
Foramen magnum	Occipital	Medulla, vertebral arteries, spinal roots of XI

Jugular foramen syndromes: Vernet IX–XI, Collet–Sicard IX–XII, Villaret IX–XII + sympathetic chain.

Paranasal sinuses compared

Sinus	At birth	Drains into	Key danger
Maxillary	Present (first)	Middle meatus (infundibulum)	Teeth roots in floor; infraorbital nerve in roof

Sinus	At birth	Drains into	Key danger
Ethmoid	Present	Anterior: middle meatus · Posterior: superior meatus	Lamina papyracea, anterior ethmoidal artery, lateral lamella
Frontal	Absent (last)	Frontal recess → middle meatus	Posterior wall to frontal lobe; Pott's puffy tumour
Sphenoid	Absent (appears 3–6 y)	Sphenoethmoidal recess	Optic nerve, ICA, cavernous sinus in lateral wall

Anterior group drains in front of the basal lamella; posterior group behind it.

Cranial fossae relations

Fossa	Floor formed by	ENT relations
Anterior	Orbital plates of frontal, cribriform plate, lesser wings	Frontal sinus, ethmoid roof, olfactory bulbs
Middle	Greater wings of sphenoid, squamous and petrous temporal	Tegmen, sphenoid sinus, cavernous sinus, foramina rotundum, ovale, spinosum
Posterior	Occipital bone, posterior petrous surface	IAM, jugular foramen, sigmoid sinus, cerebellum (mastoid relation)

Chandler classification

Stage	Name	Key signs	Usual management
I	Preseptal cellulitis	Lid oedema, normal vision and movement	Oral or IV antibiotics
II	Orbital cellulitis	Proptosis, chemosis, painful movement	IV antibiotics, nasal decongestion
III	Subperiosteal abscess	Globe displaced down and out	IV antibiotics ± drainage
IV	Orbital abscess	Severe proptosis, ophthalmoplegia, visual loss	Surgical drainage
V	Cavernous sinus thrombosis	Bilateral signs, III–VI palsies, sepsis	IV antibiotics, consider anticoagulation

Drain whenever vision is threatened, regardless of stage.

M Mnemonics

Keros types

1-3, 4-7, 8-16

Keros I 1–3 mm, II 4–7 mm, III 8–16 mm

Sinus order at birth

ME first, SF later

Maxillary and Ethmoid present; Sphenoid and Frontal develop later

Keros risk

The deeper the fossa, the thinner the wall

Keros III has the tallest, thinnest lateral lamella

Q Rapid recall

1. **Order of sinus development?** Maxillary and ethmoid present at birth; sphenoid appears around 3–6 years; frontal is last.
2. **Which teeth lie in the floor of the maxillary sinus?** Molars and the second premolar (sometimes the first premolar).
3. **Keros types by olfactory fossa depth?** I 1–3 mm, II 4–7 mm, III 8–16 mm (highest risk).

4. **What is an Onodi cell and why does it matter?** A posterior ethmoid cell lateral or above the sphenoid; the optic nerve may run in its wall.
5. **What is a Haller cell?** An infraorbital ethmoid cell along the orbital floor that can narrow the maxillary infundibulum.
6. **Structures in the lateral wall of the sphenoid sinus?** Optic nerve, ICA (with the opticocarotid recess), cavernous sinus with III, IV, V1, V2, VI.
7. **Commonest type of sphenoid pneumatisation?** Sellar (about 80%).
8. **Risk from cutting the anterior ethmoidal artery?** It retracts into the orbit causing an orbital haematoma; treat with lateral canthotomy.
9. **Contents of the pterygopalatine fossa?** Third part of the maxillary artery, maxillary nerve and pterygopalatine ganglion.
10. **Capacity of the adult maxillary sinus?** About 15 mL.
11. **Where is the maxillary sinus floor relative to the nasal floor in adults?** About 1 cm below.
12. **Frequency of accessory maxillary ostia?** About 30%.
13. **Thinnest wall of the frontal sinus?** The floor (orbital roof).
14. **What separates anterior from posterior ethmoid cells?** The basal (ground) lamella of the middle turbinate.
15. **Most anterior ethmoid cell?** Agger nasi cell.
16. **Bent–Kuhn frontal cell types?** I single cell above agger nasi; II tier of cells; III single cell into the frontal sinus; IV isolated cell within the sinus.
17. **Thinnest bone of the anterior skull base?** The lateral lamella of the cribriform plate.
18. **Where does the sphenoid ostium open?** Into the sphenoethmoidal recess, about 1–1.5 cm above the choana.
19. **What is the opticocarotid recess?** Depression in the lateral sphenoid wall between the optic nerve above and ICA below.
20. **What runs in the floor of the sphenoid sinus?** The vidian nerve in the pterygoid canal.
21. **Lund–Mackay score range?** 0–24 (each sinus 0–2, OMC 0 or 2, both sides).

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.